Android Application Development Lab

Manav Rachna International Institute of Research and Studies

School of Computer Applications

Submitted By	
Student Name	Rahul Sehraya
Roll No	25/SCA/MCAN/081
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Department	School of Computer Applications
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Submitted To	
Faculty Name	Ms. Urvashi Sangwan

MANAV RACHNA
vidyanatariksha

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ACCREDITED

Manav Rachna Campus Rd, Gadakhor Basti Village, Sector 43, Faridabad, Haryana 121004

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Linear Layout : Write a program to show the use of Linear Layout

Xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:padding="16dp">
  <!-- Top Row -->
  <LinearLayout
    android:orientation="horizontal"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:weightSum="2">
    <TextView
      android:id="@+id/tvTimer"
      android:layout_width="0dp"
      android:layout_height="wrap_content"
      android:layout_weight="1"
      android:text="Time: 30"
      android:textSize="18sp"
      android:textStyle="bold"/>
    <TextView
      android:id="@+id/tvScore"
      android:layout_width="0dp"
      android:layout_height="wrap_content"
      android:layout_weight="1"
      android:gravity="end"
      android:text="Score: 0"
      android:textSize="18sp"
      android:textStyle="bold"/>
  </LinearLayout>
  <!-- Center -->
  <LinearLayout
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    android:gravity="center">
    <TextView
      android:id="@+id/tvQuestion"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="12 + 34"
```

```
        android:textSize="28sp"
        android:layout_marginBottom="16dp"/>
    <EditText
        android:id="@+id/etAnswer"
        android:layout_width="120dp"
        android:layout_height="wrap_content"
        android:inputType="number"
        android:gravity="center"
        android:hint="Answer"/>
  </LinearLayout>
  <!-- Bottom -->
  <LinearLayout
    android:orientation="horizontal"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:weightSum="2">
    <ImageView
        android:id="@+id/imgCorrect"
        android:layout_width="0dp"
        android:layout_height="80dp"
        android:layout_weight="1"
        android:src="@android:drawable/checkbox_on_background"
        android:alpha="0.3"/>
    <ImageView
        android:id="@+id/imgWrong"
        android:layout_width="0dp"
        android:layout_height="80dp"
        android:layout_weight="1"
        android:src="@android:drawable/ic_delete"
        android:alpha="0.3"/>
  </LinearLayout>
</LinearLayout>
```

Java :

```
package com.example.linearlayout;
public class MainActivity extends AppCompatActivity {
  private TextView tvTimer, tvScore, tvQuestion;
  private EditText etAnswer;
  private ImageView imgCorrect, imgWrong;
  private int score = 0;
  private int correctAnswer = 0;
  private Random random = new Random();
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    tvTimer = findViewById(R.id.tvTimer);
    tvScore = findViewById(R.id.tvScore);
    tvQuestion = findViewById(R.id.tvQuestion);
    etAnswer = findViewById(R.id.etAnswer);
    imgCorrect = findViewById(R.id.imgCorrect);
    imgWrong = findViewById(R.id.imgWrong);
    generateQuestion();
    startTimer();
    etAnswer.setOnEditorActionListener((v, actionId, event) -> {
        checkAnswer();
        return true;
    });
}
private void startTimer() {
    new CountDownTimer(30000, 1000) {
        @Override
        public void onTick(long millisUntilFinished) {
            tvTimer.setText("Time: " + millisUntilFinished / 1000);
        }
        @Override
        public void onFinish() {
            tvTimer.setText("Time: 0");
            etAnswer.setEnabled(false);
        }
    }.start();
}
private void generateQuestion() {
    int a = random.nextInt(90) + 10; // 2-digit
    int b = random.nextInt(90) + 10;
    boolean isAddition = random.nextBoolean();
    if (isAddition) {
        tvQuestion.setText(a + " + " + b);
        correctAnswer = a + b;
    } else {
        tvQuestion.setText(a + " - " + b);
        correctAnswer = a - b;
    }
    etAnswer.setText("");
    imgCorrect.setAlpha(0.3f);
    imgWrong.setAlpha(0.3f);
}
private void checkAnswer() {
    String input = etAnswer.getText().toString();
    if (input.isEmpty()) return;
```

```
        int userAnswer = Integer.parseInt(input);
        if (userAnswer == correctAnswer) {
            score++;
            imgCorrect.setAlpha(1.0f);
            imgWrong.setAlpha(0.3f);
        } else {
            imgWrong.setAlpha(1.0f);
            imgCorrect.setAlpha(0.3f);
        }

        tvScore.setText("Score: " + score);
        generateQuestion();
    }
}
```

Time: 10
Score: 1
31 - 63
Answer
1:1

Relative Layout : Write a program to show the use of Relative Layout

Xml :

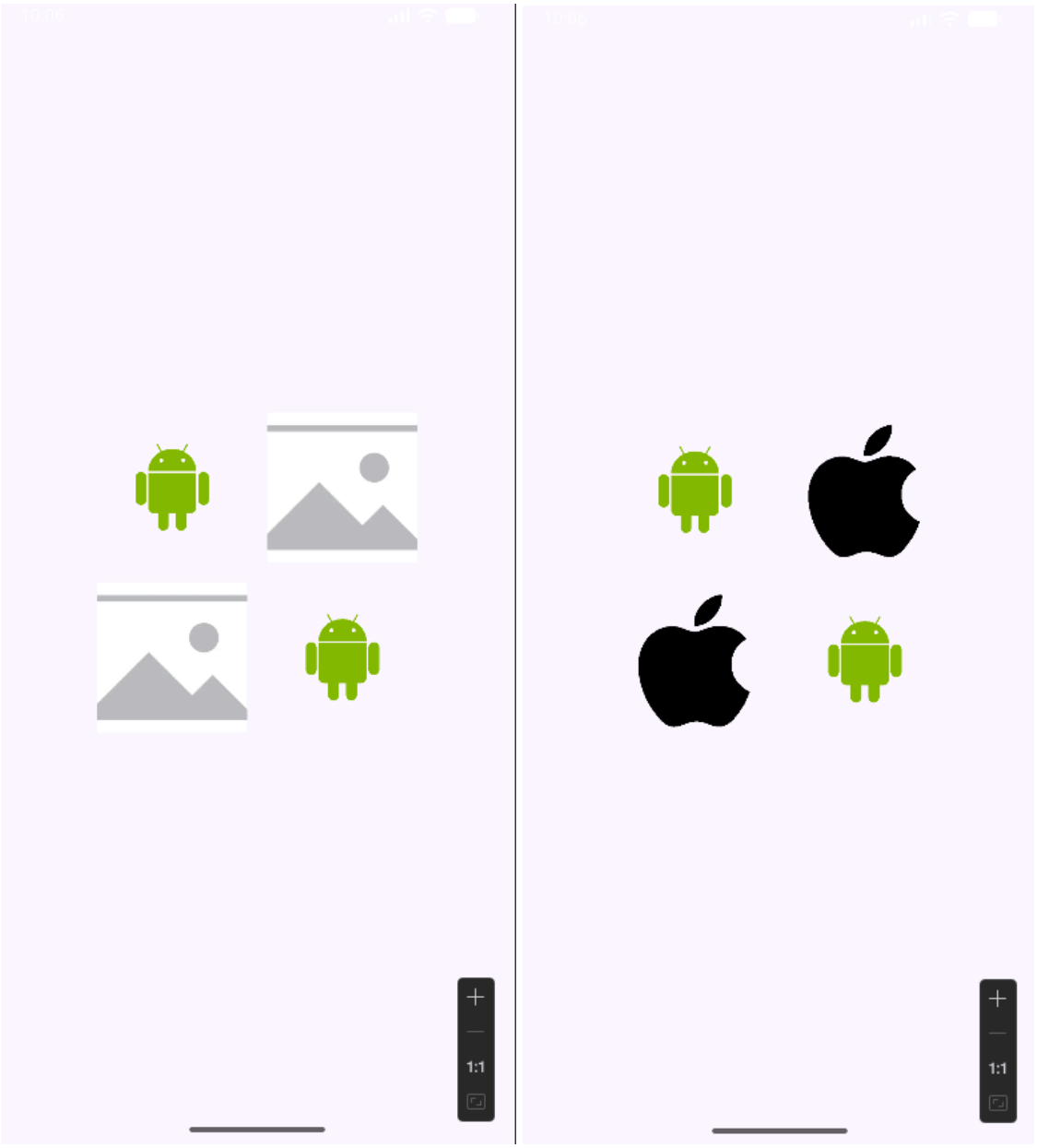
```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:padding="16dp">

  <RelativeLayout
     android:id="@+id/cardContainer"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:layout_centerInParent="true">

     <ImageView
        android:id="@+id/card1"
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:src="@drawable/img1"
        android:scaleType="centerCrop" />

     <ImageView
        android:id="@+id/card2"
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:layout_toEndOf="@id/card1"
        android:layout_alignTop="@id/card1"
        android:layout_marginStart="16dp"
        android:src="@drawable/img1"
        android:scaleType="centerCrop" />

           <ImageView
        android:id="@+id/card3"
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:layout_below="@id/card1"
        android:layout_alignStart="@id/card1"
        android:layout_marginTop="16dp"
        android:src="@drawable/img1"
        android:scaleType="centerCrop" />
```

```
    <ImageView
        android:id="@+id/card4"
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:layout_below="@id/card2"
        android:layout_alignStart="@id/card2"
        android:layout_marginTop="16dp"
        android:src="@drawable/img1"
        android:scaleType="centerCrop" />

  </RelativeLayout>

</RelativeLayout>
```

Java :

```
public class MainActivity extends AppCompatActivity {

  ImageView card1, card2, card3, card4;
  ImageView firstCard, secondCard;

  int firstImage, secondImage;
  boolean isBusy = false;

  Integer[] images = {
        R.drawable.android,
        R.drawable.android,
        R.drawable.apple,
        R.drawable.apple
  };

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    card1 = findViewById(R.id.card1);
    card2 = findViewById(R.id.card2);
    card3 = findViewById(R.id.card3);
    card4 = findViewById(R.id.card4);

    List<Integer> imageList = Arrays.asList(images);
    Collections.shuffle(imageList);

    card1.setTag(imageList.get(0));
    card2.setTag(imageList.get(1));
```

```
        card3.setTag(imageList.get(2));
        card4.setTag(imageList.get(3));

        View.OnClickListener clickListener = view -> {
            if (isBusy) return;

            ImageView card = (ImageView) view;

            if (card == firstCard) return;

            int image = (int) card.getTag();
            card.setImageResource(image);

            if (firstCard == null) {
                firstCard = card;
                firstImage = image;
            } else {
                secondCard = card;
                secondImage = image;
                isBusy = true;
                checkMatch();
            }
        };

        card1.setOnClickListener(clickListener);
        card2.setOnClickListener(clickListener);
        card3.setOnClickListener(clickListener);
        card4.setOnClickListener(clickListener);
    }

    private void checkMatch() {
        new Handler().postDelayed(() -> {

            if (firstImage == secondImage) {
                firstCard.setEnabled(false);
                secondCard.setEnabled(false);
            } else {
                firstCard.setImageResource(R.drawable.img1);
                    secondCard.setImageResource(R.drawable.img1);
            }

            firstCard = null;
            secondCard = null;
            isBusy = false;

        }, 800);
    }
}
```


10:06
1:1
10:06
1:1

Grid Layout : Write a program to show the use of Grid Layout

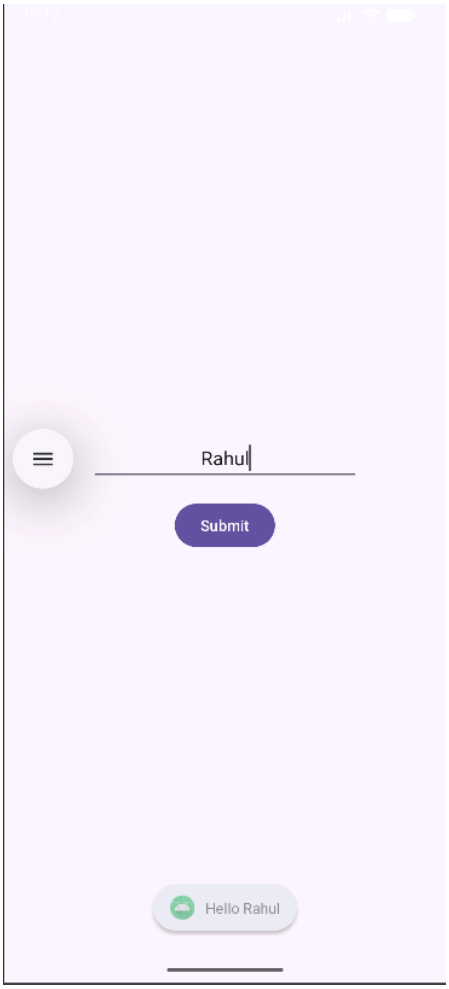
Xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="20dp">

  <EditText
     android:id="@+id/etName"
     android:layout_width="250dp"
     android:layout_height="wrap_content"
     android:hint="@string/enter_your_name"
     android:gravity="center"/>

  <Button
     android:id="@+id/btnSubmit"
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="@string/submit"
     android:layout_marginTop="16dp"/>
</LinearLayout>
```

Java :

```
public class MainActivity extends AppCompatActivity {

  EditText etName;
  Button btnSubmit;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_main);

     etName = findViewById(R.id.etName);
     btnSubmit = findViewById(R.id.btnSubmit);

     btnSubmit.setOnClickListener(new View.OnClickListener() {
```

```
        @Override
        public void onClick(View view) {
            String name = etName.getText().toString();

            Toast.makeText(MainActivity.this,
                    "Hello " + name,
                    Toast.LENGTH_SHORT).show();
        }
    });
  }
}
```



Constraint Layout : Write a program to show the use of Constraint Layout

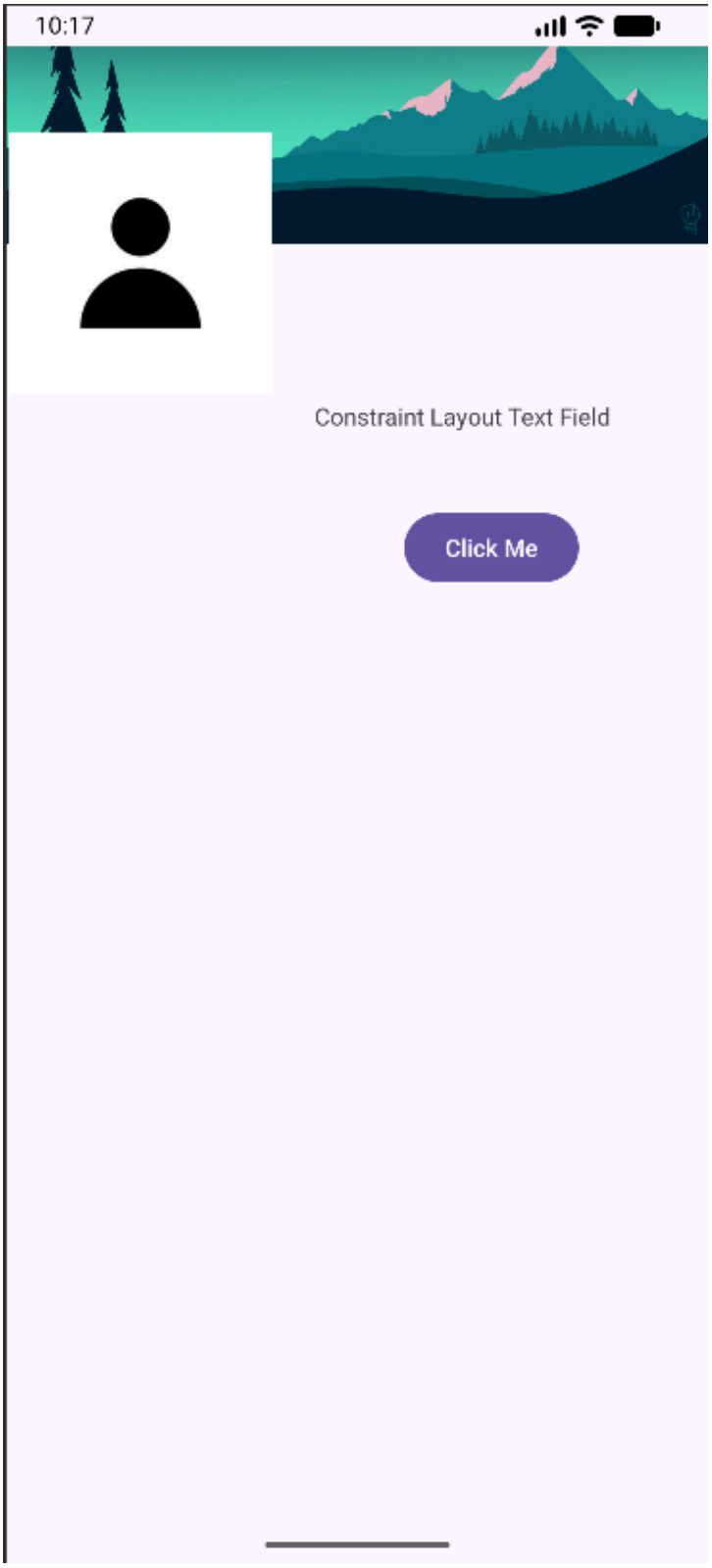
Xml :

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">

  <ImageView
    android:id="@+id/imageView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    app:layout_constraintBottom_toTopOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:srcCompat="@drawable/bgimg" />

  <ImageView
    android:id="@+id/imageView2"
    android:layout_width="157dp"
    android:layout_height="154dp"
    app:layout_constraintBottom_toBottomOf="@+id/imageView"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.0"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.179"
    app:srcCompat="@drawable/pfp" />

  <Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Click Me"
    app:layout_constraintBottom_toBottomOf="@+id/imageView"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toEndOf="@+id/imageView2"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.7" />
```

```
  <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Constraint Layout Text Field"
      app:layout_constraintBottom_toBottomOf="@+id/imageView"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintHorizontal_bias="0.76"
      app:layout_constraintStart_toStartOf="@+id/imageView2"
      app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Java :

```
public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);
      ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
          Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
          v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
          return insets;
      });
  }
}
```


10:17
Constraint Layout Text Field
Click Me

Menus : Write a program to show the use of Menus using MenuInflater

Xml :

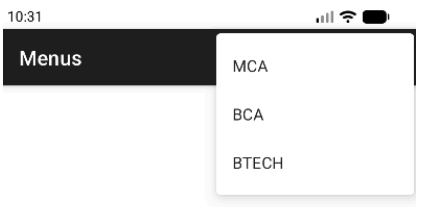
```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
  <item android:id="@+id/mca"
      android:title="MCA"
      />

  <item android:id="@+id/bca"
      android:title="BCA"

      />
  <item android:id="@+id/btech"
      android:title="BTECH"
      />
</menu>
```

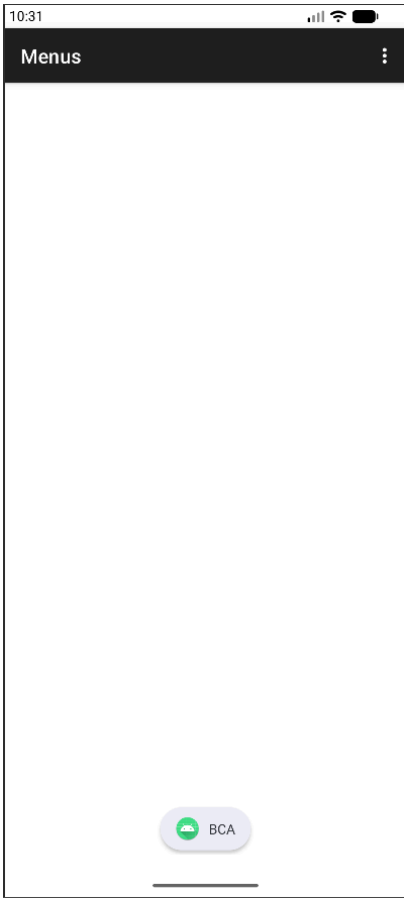
Java :

```
public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);
/*        Toolbar toolbar = findViewById(R.id.toolbar);
      setSupportActionBar(toolbar);*/
  }
  @Override
  public boolean onCreateOptionsMenu(Menu menu) {
      getMenuInflater().inflate(R.menu.menu_main, menu);
      return true;
  }

  @Override
  public boolean onOptionsItemSelected(@NonNull MenuItem item) {
      int id = item.getItemId();
      if (id == R.id.mca) {
          Toast.makeText(this, "MCA", Toast.LENGTH_SHORT).show();
          return true;
      } else if (id == R.id.bca) {
```

```
            Toast.makeText(this, "BCA", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.btech) {
            Toast.makeText(this, "BTECH", Toast.LENGTH_SHORT).show();
            return true;
        }
        return super.onOptionsItemSelected(item);
    }
}
```

10:30
Menus


10:31
Menus
MCA
BCA
BTECH


10:31
Menus
BCA

Intents : Create an activity to navigate to different activities using Intents

Xml :

Page 1:

```
<TextView
  android:id="@+id/textView"
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:text="Main Page "
  android:textAlignment="center"
  android:textSize="20sp"
  android:textStyle="bold"
  app:layout_constraintBottom_toBottomOf="parent"
  app:layout_constraintEnd_toEndOf="parent"
  app:layout_constraintStart_toStartOf="parent"
  app:layout_constraintTop_toTopOf="parent"
  app:layout_constraintVertical_bias="0.153" />

<Button
  android:id="@+id/btn"
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:text="Go To Page 2"
  app:layout_constraintBottom_toBottomOf="parent"
  app:layout_constraintEnd_toEndOf="parent"
  app:layout_constraintHorizontal_bias="0.51"
  app:layout_constraintStart_toStartOf="parent"
  app:layout_constraintTop_toBottomOf="@+id/textView"
  app:layout_constraintVertical_bias="0.312" />
```

Page 2 :

```
<Button
  android:id="@+id/button"
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:text="Go Back to Page 1"
  app:layout_constraintBottom_toBottomOf="parent"
  app:layout_constraintEnd_toEndOf="parent"
  app:layout_constraintStart_toStartOf="parent"
  app:layout_constraintTop_toTopOf="parent"
  app:layout_constraintVertical_bias="0.368" />

<TextView
```

```
  android:id="@+id/textView2"
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:text="Page 2"
  android:textSize="24sp"
  app:layout_constraintBottom_toBottomOf="parent"
  app:layout_constraintEnd_toEndOf="parent"
  app:layout_constraintHorizontal_bias="0.472"
  app:layout_constraintStart_toStartOf="parent"
  app:layout_constraintTop_toTopOf="parent"
  app:layout_constraintVertical_bias="0.164" />

<Button
  android:id="@+id/button2"
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:text="External Connect"
  app:layout_constraintBottom_toBottomOf="parent"
  app:layout_constraintEnd_toEndOf="parent"
  app:layout_constraintStart_toStartOf="parent"
  app:layout_constraintTop_toTopOf="parent" />
```

Java :

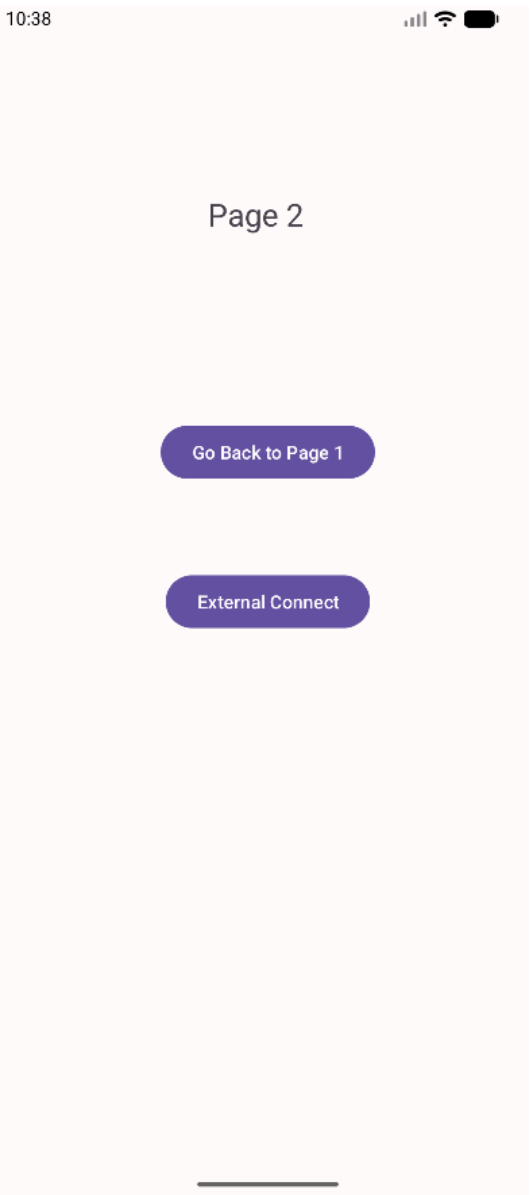
Page 1:

```
public class MainActivity extends AppCompatActivity {
  Button b1;
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);
    b1 = findViewById(R.id.btn);
    b1.setOnClickListener(new View.OnClickListener() {
      @Override
      public void onClick(View v) {
        Intent i = new Intent(MainActivity.this, Page2.class);
        startActivity(i);
      }
    });
  }
}
```

Page 2 :

```
public class Page2 extends AppCompatActivity {
```

```
    Button b1;
    Button b2;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_page2);

        b1 = findViewById(R.id.button);
        b2 = findViewById(R.id.button2);

        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent(Page2.this, MainActivity.class);
                startActivity(intent);
            }
        });

        b2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent(Intent.ACTION_VIEW, Uri.parse("http://www.google.com"));
                startActivity(intent);
            }
        });

    }
}
```



Rahul Sehraya (25/SCA/MCAN/081)

MP3 Player: Create a program that plays, pauses and resumes mp3 files

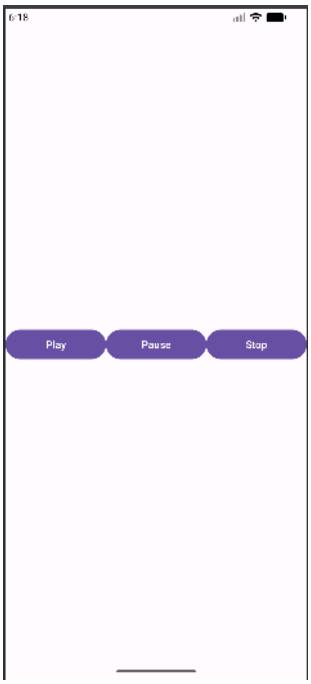
Xml :

```
<GridLayout
  android:layout_width="match_parent"
  android:layout_height="wrap_content"
  android:columnCount="3">

  <Button
     android:id="@+id/playbtn"
     android:layout_width="0dp"
     android:layout_height="wrap_content"
     android:layout_columnWeight="1"
     android:text="Play"/>

  <Button
     android:id="@+id/pausebtn"
     android:layout_width="0dp"
     android:layout_height="wrap_content"
     android:layout_columnWeight="1"
     android:text="Pause"/>

  <Button
     android:id="@+id/stopbtn"
     android:layout_width="0dp"
     android:layout_height="wrap_content"
     android:layout_columnWeight="1"
     android:text="Stop"/>
</GridLayout>
```

Java :

```
public class MainActivity extends AppCompatActivity {
  MediaPlayer player;
  Button playbtn, pausebtn, stopbtn;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     EdgeToEdge.enable(this);
     setContentView(R.layout.activity_main);

     player = MediaPlayer.create(this, R.raw.music);
```

```
        playbtn  = findViewById(R.id.playbtn);
        pausebtn = findViewById(R.id.pausebtn);
        stopbtn  = findViewById(R.id.stopbtn);

        playbtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                player.start();
            }
        });

        pausebtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                player.pause();
            }
        });

        stopbtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                player.stop();
                player.reset();
                player.release();
                player = MediaPlayer.create(MainActivity.this, R.raw.music);
            }
        });
    }
}
```



Alarm : Create an alarm that rings every Sunday at 8:00 AM. Modify it to use a time picker to set alarm time

Xml :

```
<TimePicker
  android:id="@+id/timePicker"
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:timePickerMode="spinner"
  app:layout_constraintBottom_toTopOf="@+id/btnSetAlarm"
  app:layout_constraintEnd_toEndOf="parent"
  app:layout_constraintStart_toStartOf="parent"
  app:layout_constraintTop_toTopOf="parent"
  app:layout_constraintVertical_chainStyle="packed" />

<Button
  android:id="@+id/btnSetAlarm"
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:layout_marginTop="24dp"
  android:text="Set Alarm"
  app:layout_constraintBottom_toBottomOf="parent"
  app:layout_constraintEnd_toEndOf="parent"
  app:layout_constraintStart_toStartOf="parent"
  app:layout_constraintTop_toBottomOf="@+id/timePicker" />
```

Java :

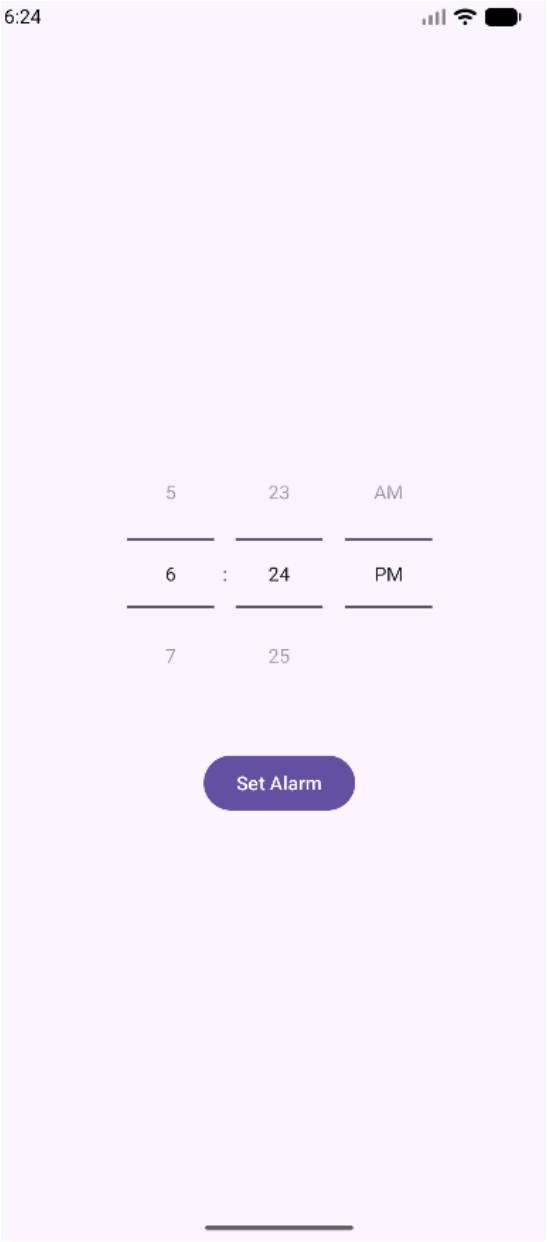
Java 1 :

```
public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);

    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
      Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
      v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
      return insets;
```

```
    });

    TimePicker timePicker = findViewById(R.id.timePicker);
    Button btnSetAlarm = findViewById(R.id.btnSetAlarm);

    btnSetAlarm.setOnClickListener(v -> {
        AlarmManager alarmManager = (AlarmManager) getSystemService(ALARM_SERVICE);

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.S) {
            if (alarmManager != null && !alarmManager.canScheduleExactAlarms()) {
                Intent intent = new Intent(Settings.ACTION_REQUEST_SCHEDULE_EXACT_ALARM);
                startActivity(intent);
                return;
            }
        }

        setAlarm(timePicker, alarmManager);
    });
}

@SuppressLint("ScheduleExactAlarm")
private void setAlarm(TimePicker timePicker, AlarmManager alarmManager) {
    int hour = timePicker.getHour();
    int minute = timePicker.getMinute();

    Calendar calendar = Calendar.getInstance();
    calendar.set(Calendar.HOUR_OF_DAY, hour);
    calendar.set(Calendar.MINUTE, minute);
    calendar.set(Calendar.SECOND, 0);

    if (calendar.getTimeInMillis() <= System.currentTimeMillis()) {
        calendar.add(Calendar.DAY_OF_MONTH, 1);
    }

    Intent intent = new Intent(this, AlarmReceiver.class);
    PendingIntent pendingIntent = PendingIntent.getBroadcast(
            this, 0, intent, PendingIntent.FLAG_IMMUTABLE
    );

    if (alarmManager != null) {
        PendingIntent showIntent = PendingIntent.getActivity(
                this, 0, new Intent(this, MainActivity.class), PendingIntent.FLAG_IMMUTABLE
        );

        AlarmManager.AlarmClockInfo alarmClockInfo = new AlarmManager.AlarmClockInfo(
                calendar.getTimeInMillis(),
                showIntent
```

```
        );

        alarmManager.setAlarmClock(alarmClockInfo, pendingIntent);

        String time = String.format(Locale.getDefault(), "%02d:%02d", hour, minute);
        Toast.makeText(this, "Alarm set for " + time, Toast.LENGTH_LONG).show();
    }
  }
}
```

Java 2:

```
public class AlarmReceiver extends BroadcastReceiver {
  @Override
  public void onReceive(Context context, Intent intent) {
    // Show immediate feedback
    Toast.makeText(context, "Alarm is ringing!", Toast.LENGTH_LONG).show();

    // Play the alarm sound
    MediaPlayer mp = MediaPlayer.create(context, R.raw.alarm_sound);
    if (mp != null) {
      mp.start();
      mp.setOnCompletionListener(MediaPlayer::release);
    }

    showNotification(context);
  }

  private void showNotification(Context context) {
    String channelId = "alarm_channel";
                        NotificationManager notificationManager = (NotificationManager)
context.getSystemService(Context.NOTIFICATION_SERVICE);

    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
      NotificationChannel channel = new NotificationChannel(
          channelId, "Alarm Channel", NotificationManager.IMPORTANCE_HIGH);
      notificationManager.createNotificationChannel(channel);
    }

    NotificationCompat.Builder builder = new NotificationCompat.Builder(context, channelId)
        .setSmallIcon(android.R.drawable.ic_lock_idle_alarm)
        .setContentTitle("Alarm")
        .setContentText("Your scheduled alarm is ringing!")
        .setPriority(NotificationCompat.PRIORITY_HIGH)
        .setAutoCancel(true);
```

```
        if (notificationManager != null) {
            notificationManager.notify(1, builder.build());
        }
    }
}
```